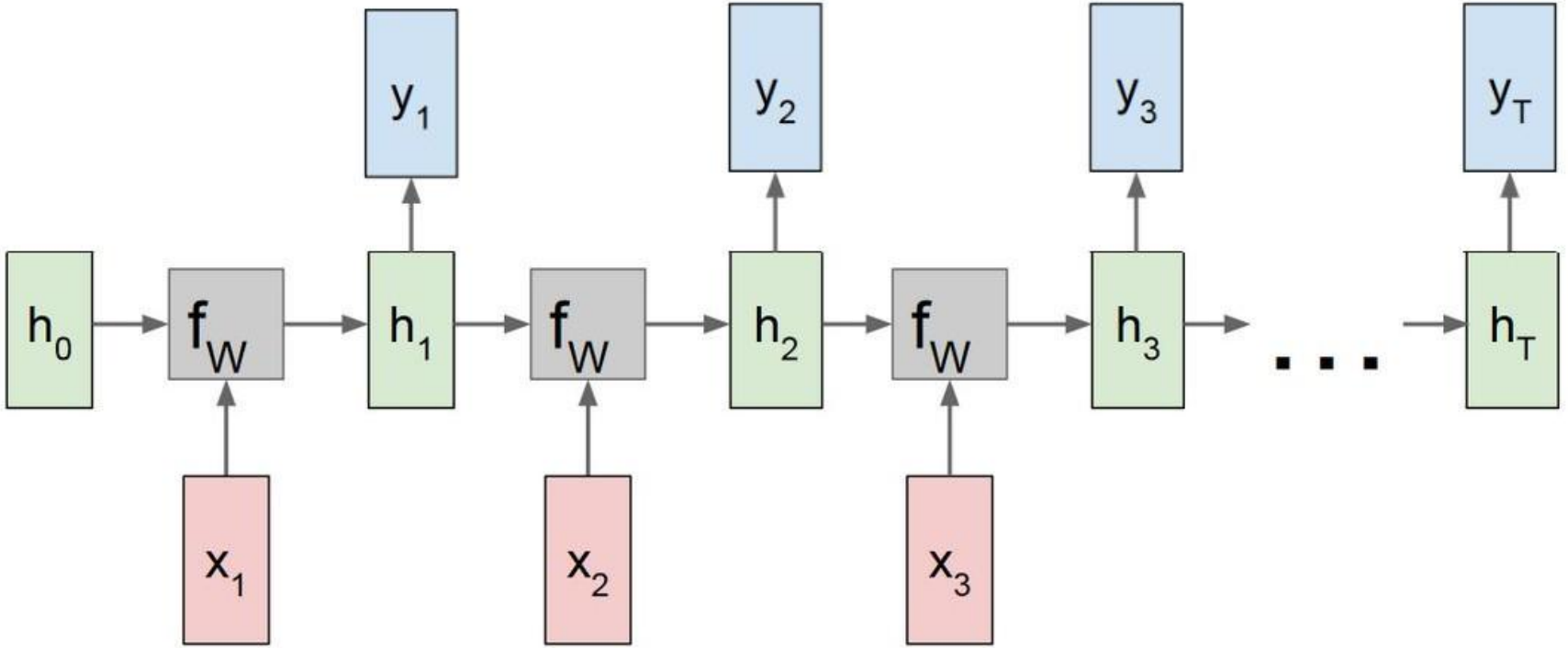
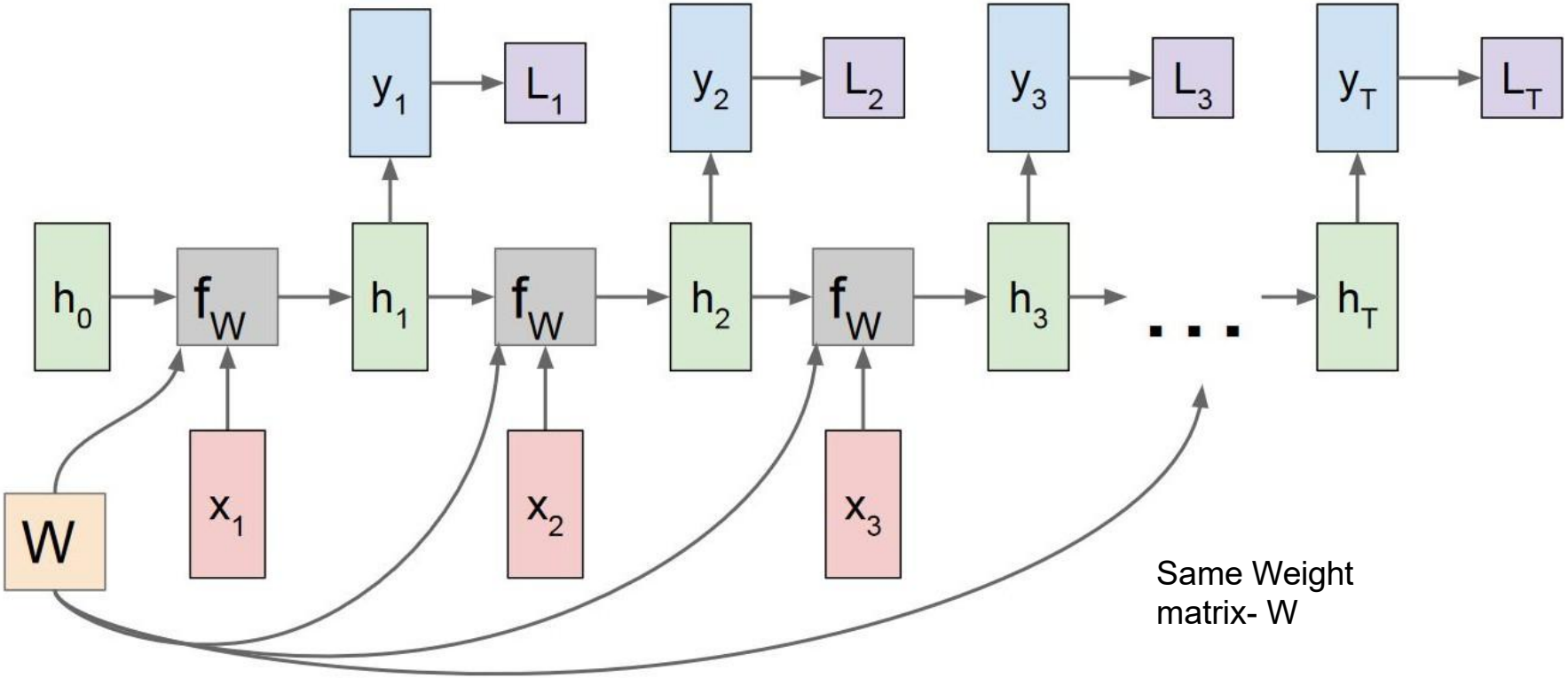
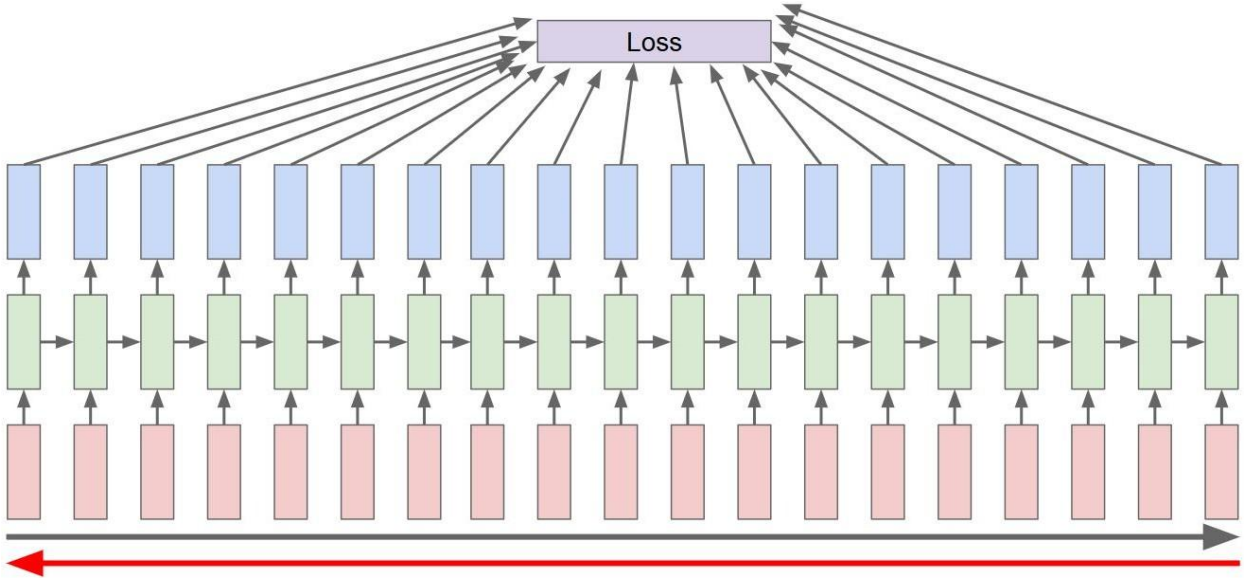
RNN- Rolled out representation



RNN- Rolled out representation

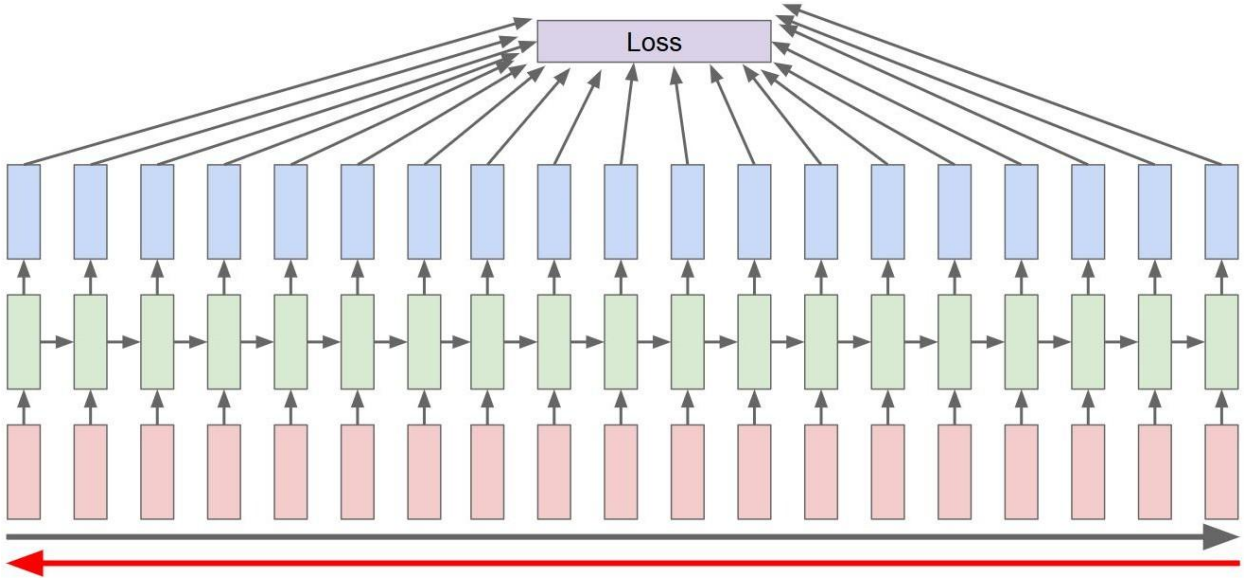


RNN- Backpropagation Through Time



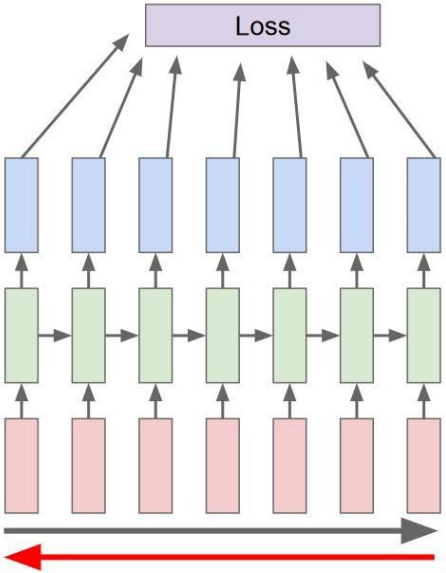
Forward pass through entire sequence to produce intermediate hidden states, output sequence and finally the loss. Backward pass through the entire sequence to compute gradient.

RNN- Backpropagation Through Time



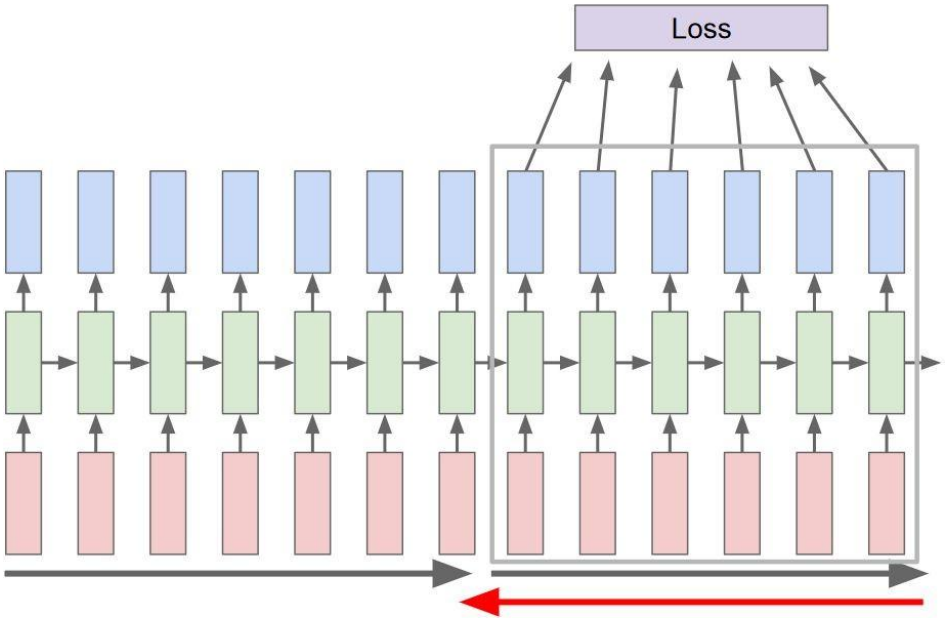
Running Backpropagation through time for the entire text would be very slow. Switch to an approximation- Truncated Backpropagation Through Time

RNN- Truncated Backpropagation Through Time



Run forward and backward through chunks of the sequence instead of whole sequence

RNN- Truncated Backpropagation Through Time



Carry hidden states forward in time forever, but only backpropagate for some smaller number of steps

RNN- Types

The 3 most common types of Recurrent Neural Networks are-

1. Vanilla RNN
2. LSTM (Long Short-Term Memory)
3. GRU (Gated Recurrent Units)

Some good resources-
Understanding LSTM Networks

An Empirical Exploration of Recurrent Network Architectures

Recurrent Neural Network Tutorial, Part 4 – Implementing a GRU/LSTM RNN with Python and Theano

Stanford CS231n: Lecture 10 | Recurrent Neural Networks

CNNs

Some slides borrowed from Fei-Fei Li & Justin Johnson & Serena Yeung at Stanford.